**EXPERIMENT :-8**

**CODE:-**

```
#include <stdio.h>

int max(int a, int b) {
   return (a > b) ? a : b;
}

int main() {
   int n, W, i, w;

   printf("Enter number of items: ");
   scanf("%d", &n);

   int weight[n], profit[n];

   printf("Enter weights:\n");
   for (i = 0; i < n; i++)
      scanf("%d", &weight[i]);

   printf("Enter profits:\n");
   for (i = 0; i < n; i++)
      scanf("%d", &profit[i]);

   printf("Enter knapsack capacity: ");
   scanf("%d", &W);
```

```
    int dp[n + 1][W + 1];

    for (i = 0; i <= n; i++) {
        for (w = 0; w <= W; w++) {

            if (i == 0 || w == 0)
                dp[i][w] = 0;

            else if (weight[i - 1] <= w)
                dp[i][w] = max(
                    profit[i - 1] + dp[i - 1][w - weight[i - 1]],
                    dp[i - 1][w]
                );

            else
                dp[i][w] = dp[i - 1][w];
        }
    }

    printf("\nMaximum Profit = %d\n", dp[n][W]);

    return 0;
}
```

## ⭐ Example Input

```
Enter number of items: 4

Weights:
2 1 3 2

Profits:
12 10 20 15

Capacity:
5
```

## Output

```
Maximum Profit = 37
```

The selected items can be:

```
Item 1 → Weight 2, Profit 12
Item 2 → Weight 1, Profit 10
Item 4 → Weight 2, Profit 15

Total Weight  = 5
Total Profit  = 37
```